System Validation Report

Test Plan VPS-TP-01 executed against the final build | 6 November 2026

1. Summary

17 system test cases: 16 passed, 1 not run, 0 failed. Each case is automated step by step in tests/test_plan_cases.py and was also walked through in the running application. Environment: Windows 11, Python 3.14, pytest 9 with pytest-cov, run on 6 November 2026 against the main branch. The same suite runs on every pull request in GitHub Actions on Ubuntu, Windows and macOS with Python 3.10 and 3.12.

2. Requirement validation

Area	Requirements	Test cases	Result
Slots and categories	REQ-SLOT-01 to 04	TC-01, 03, 05, 10	Validated
Entry and tickets	REQ-LOG-01 to 04	TC-01, 02	Validated
Fees and payment	REQ-PAY-01 to 03, BR-01, BR-02	TC-06, 07	Validated
Reports	REQ-REP-01 to 03	TC-08	Validated
Users and audit	REQ-USR-01, 02	TC-15	Validated
Smart assignment and route	REQ-NAV-01 to 04	TC-04	Validated
Prediction and demo data	REQ-PRED-01 to 04	TC-05, 09	Validated
Performance	NFR-PERF-01 to 05	TC-04, 08, 10, 11	Validated on reference machine
Reliability	NFR-REL-01 to 04	TC-02, 07, 11, 12	Validated
Security	SEC-01 to 08	TC-14 to 17	Validated
Usability	QA-USE-01	TC-13	Not yet validated
Portability	QA-POR-01, OR-02	TC-12	Validated (Windows by hand, Ubuntu and macOS by CI)
Maintainability and modularity	QA-MNT-01, QA-MOD-01	Inspection	MNT met; MOD partly met (D-13)
Business and other rules	BR-03, OR-01, OR-03, OR-04	TC-09	Validated

3. Test case results

TC	Requirements	What was checked	Actual result	Status
TC-01	LOG-01..03, SLOT-01, NAV-02	"TN 09 AB 1234", Car, all slots vacant	Stored as TN09AB1234, ticket VPS-YYYYMMDD-NNNN, QR served, S01 Occupied, entry time to the second	Pass
TC-02	LOG-01, LOG-04, REL-01	"", "AB1", 12 characters, "TN-09@AB"; repeat active plate	HTTP 400 with a clear message and no ticket; duplicate HTTP 409; no stack trace	Pass
TC-03	SLOT-03	EV, Two-Wheeler, Car with Accessible	S17, S11, S15	Pass
TC-04	NAV-01..04	S01, S02 occupied; route; exit mode; target taken	S03 with path 4; S10 in exit mode; next best slot assigned in under 500 ms	Pass
TC-05	SLOT-04, PRED-01..03	Car slots full, average 90 min, longest parked 86 min	Refused: "No Car slot available", "expected in about 4 min"	Pass
TC-06	PAY-01, BR-01, BR-02	Car 2 h 10 min; EV 1 h 0 min 1 s; Car 4:59, 5:00, 5:01	Rs. 120; Rs. 100; Rs. 0, Rs. 0, Rs. 40	Pass
TC-07	PAY-02, 03, REL-04	Pay with force failure on, then off	HTTP 402, ticket active, slot occupied; then receipt, slot Vacant, gate OPEN	Pass
TC-08	REP-01..03, PERF-05	Daily, weekly, monthly over 30 days; CSV; PDF	Totals equal SQL SUM; peak hour shown; exports generated; report in 41 ms	Pass
TC-09	PRED-04, OR-01, BR-03, OR-04	Fresh DB; seed; clock to 23 h 55 m, then 24 h 5 m	20 slots, 2 accounts, 200 records; Overstay only after 24 h	Pass

TC	Requirements	What was checked	Actual result	Status
TC-10	PERF-01, 02, 04, SLOT-02	20 entries and exits; BFS 100 runs on 100 cells	P95 23 ms; BFS max 0.33 ms; grid polls every 500 ms	Pass
TC-11	PERF-03, REL-02	Parallel entries, then 5 for 1 free slot	All distinct slots; exactly 1 success, 4 refused; no duplicate active slot	Pass
TC-12	REL-03, OR-02, QA-POR-01	Restart with active ticket; other operating systems	No data lost. Windows run by hand; Ubuntu and macOS covered by the CI test run, not by a manual install	Pass (CI for Ubuntu and macOS)
TC-13	QA-USE-01	5 first-time users, entry to receipt, unaided	Needs real users; planned for the testing sprint	Not run
TC-14	SEC-01, 02	Users table; 5 wrong passwords; correct one; 5 min later	Distinct scrypt hashes; locked and logged; unlocks after 5 min (Demo Clock)	Pass
TC-15	SEC-03, 08, USR-01, 02	Attendant on admin routes; admin rate change; audit DELETE	403 and logged; old and new values audited; DELETE returns 405	Pass
TC-16	SEC-04, 05	SQL injection strings in plate, ticket, login; script tag	No bypass or database change; script shown as escaped text	Pass
TC-17	SEC-06, 07	Cookie flags; idle 16 min; replay after logout; bad CSRF tokens	HttpOnly and SameSite=Lax; redirected to login; old cookie refused; HTTP 400	Pass

4. Open items

TC-13 (5 first-time users completing entry to receipt within 3 minutes) needs real participants and has not been run. Defects found during validation were fixed before this report: the layout and demo passwords differed from the Test Plan (#37), a second login ended the first session (#26), and timing checks were unreliable on shared CI runners (#47).